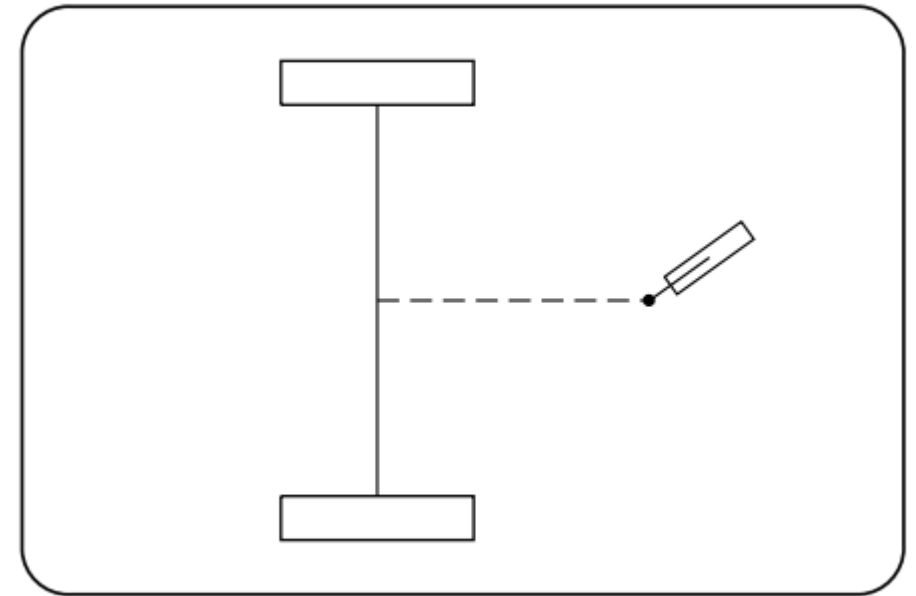




Differential drive vehicle

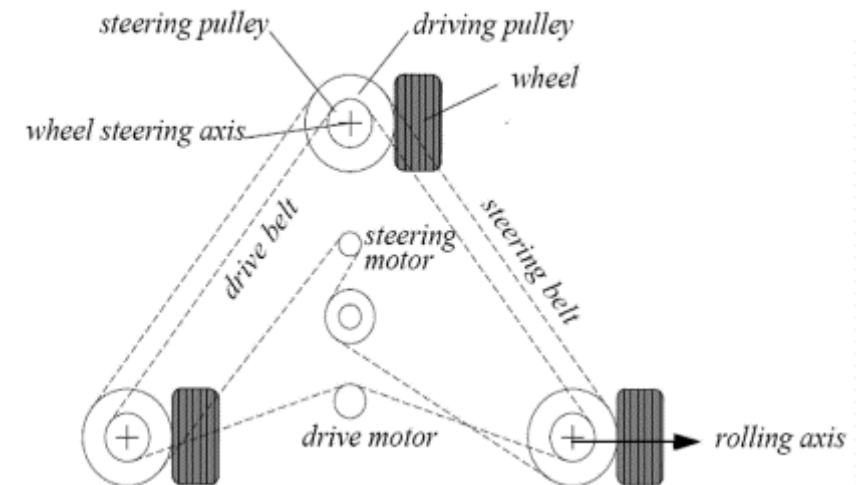
- In a differential drive robot there are
 - Two actuated fixed wheels with the same axis
 - A caster wheel to keep the robot statically balanced
 - The robot can rotate by applying a different velocity to the wheels, or move straight applying the same angular velocity
 - It can rotate on the spot by setting the two speeds to opposite values





Synchro Drive robots

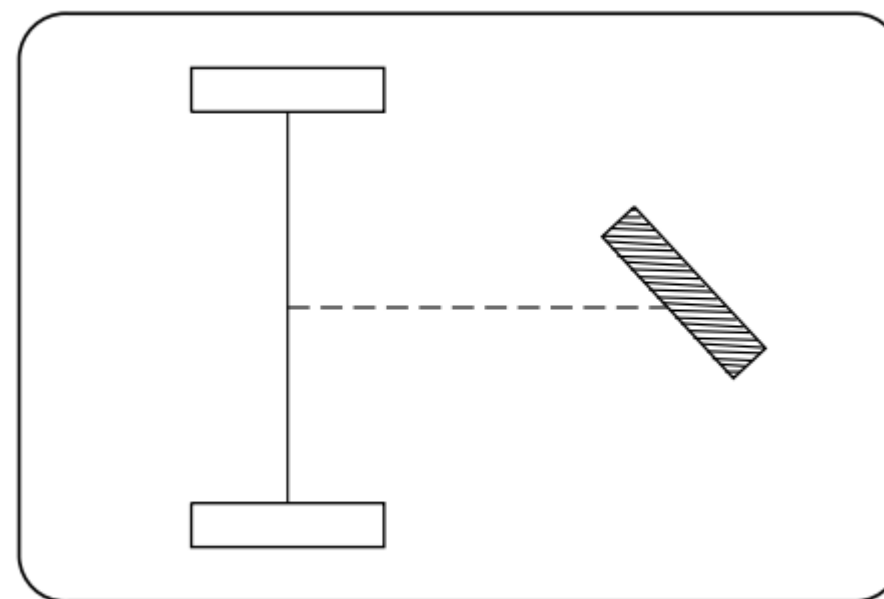
- In the synchro drive configuration there are three steerable wheels connected by a chain
- There are two motors: one rotates the three wheels “synchronously” and the other transmits the motion (synchronously) to all of them
- It can do the same kinematics of the differential drive but it needs only one motor to move straight





Tricycle robots

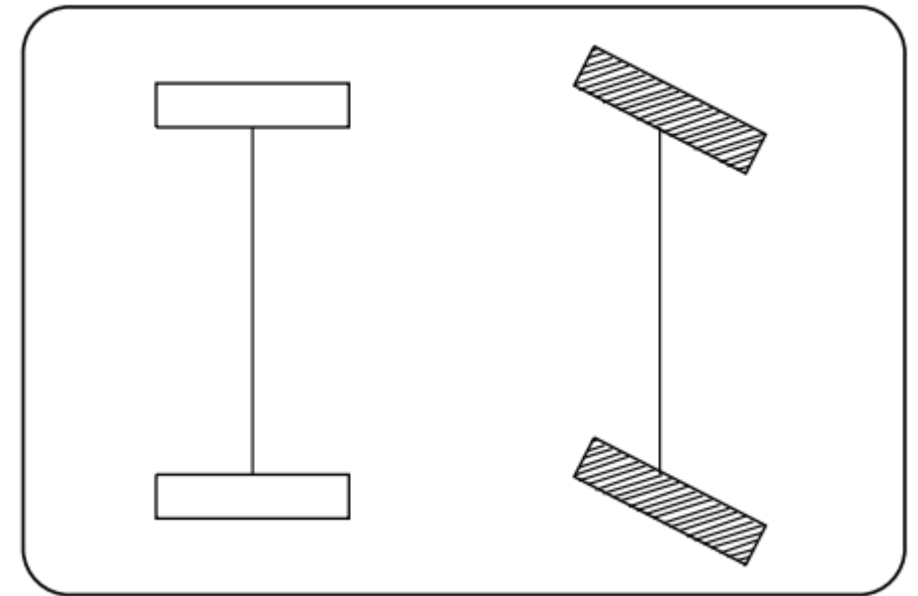
- In a tricycle we have two wheels that are fixed and actuated by a motor
- The third wheel is steerable and its rotation around the vertical axis is governed by another motor
- It is also possible that the two motors operate both on the steering wheel to turn it and to secure locomotion





Carlike

- The carlike robot is very similar to a tricycle except that it has two turning wheels
- Again the traction can be on the fixed wheels (rear-wheel traction) or on the front wheel (front-wheel traction)
- Contrary to a differential drive robot, a car-like (and a tricycle) cannot turn on the spot and have a limited curvature radius
- Besides, the carlike has an interesting problem





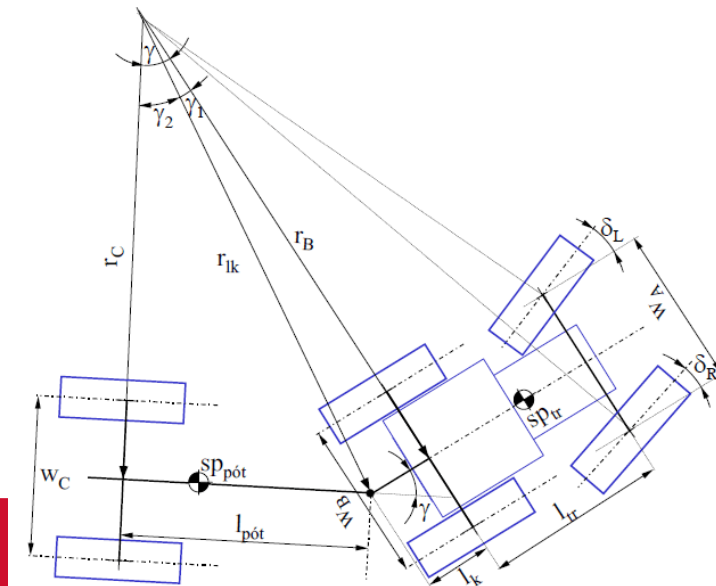
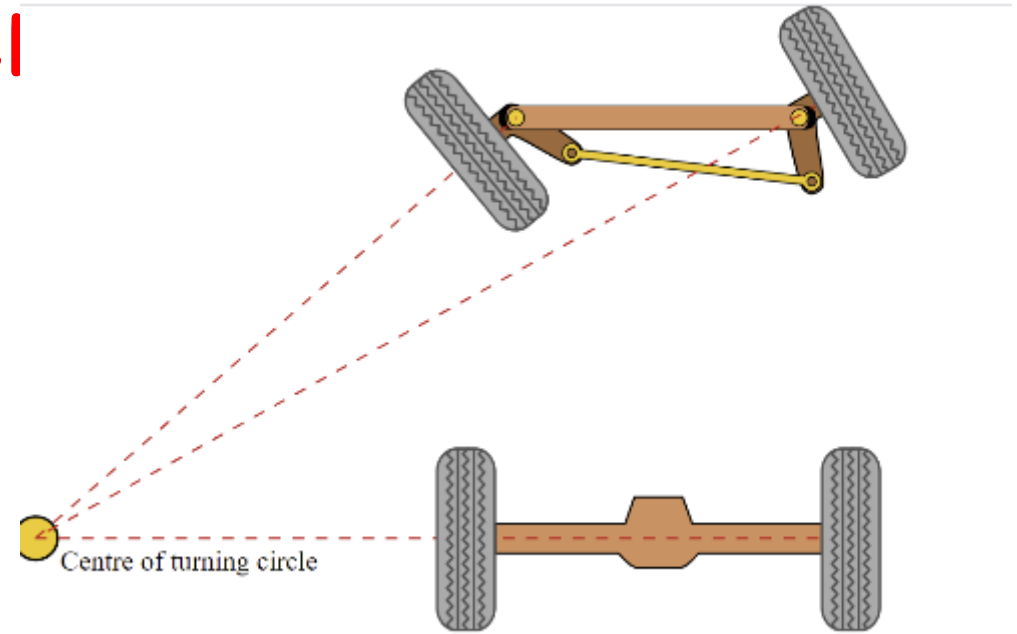
Ackerman steering

- Back in 1758 Erasmus Darwin was driving his horse-drawn carriage.
- The two front wheels turned of the same angle
- One of the wheels (the external one) slipped sideways and it caused the carriage to tip over his driver
- The solution was devised by Lankensperger in Munich and patented by his English agent (Ackerman) in London
- The idea is very simple: the two wheels have to follow the same circular trajectory



Ackerman Steer

- In order to follow the same circle the internal wheel has to turn more than the external one
- This is solved, in modern cars, by the geometry of the axle
- The Ackerman steering can be generalised to the case of trailers.





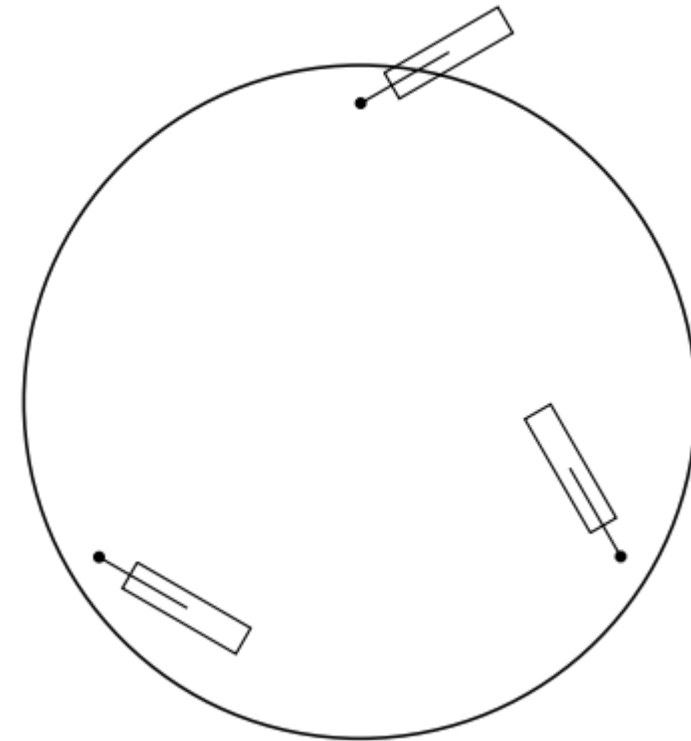
Constraints

- Contrary the to manipulators mobile robots do not usually have a limited workspace
 - They can reach any location in the Euclidean space
- However they do have motion constraints
 - A Differential Drive robot is quite flexible, but it cannot move sideways along the axis connecting the actuated wheels
 - This is called a nonholonomic constraint
 - In addition, a carlike has a limited curvature radius



An exception

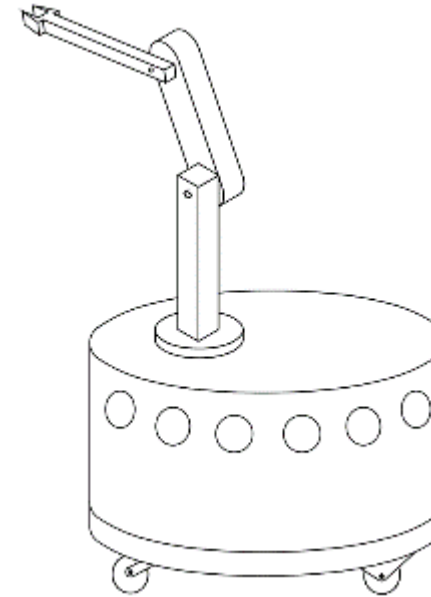
- An exception is an omnibot
- An omnibot is characterised by three autonomously actuated caster-wheels (usually in symmetric positions)
- It can move in *any direction*





Additional types of robots

- Clearly, we can mount an anthropomorphic manipulator and a differential drive mobile base in order to improve its working abilities
- We can improve the dexterity by additional degrees of freedom (7 for the Kuka LWR)





More sophisticated end-effectors

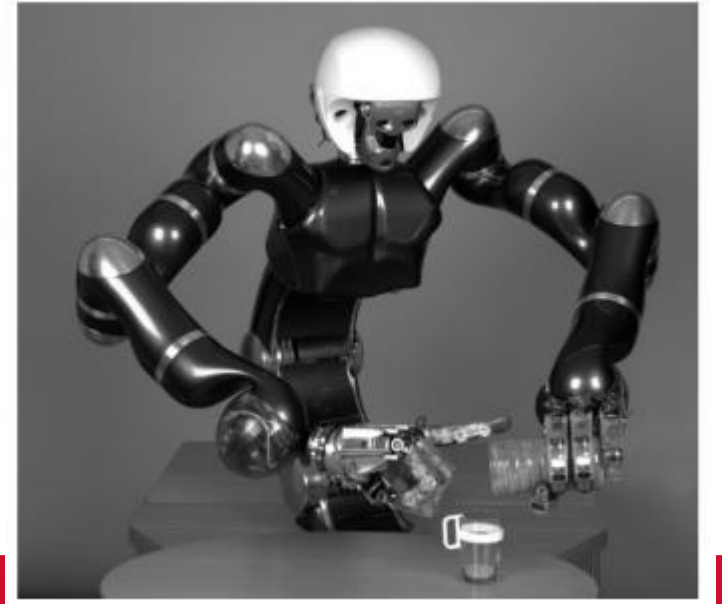
- A very active research area has been on the development of sophisticated anthropomorphic hands
- Such hands can be used for
 - rehabilitation and assistive purposes (e.g., as a replacement for a severed human hands)
 - Creation of sophisticated anthropomorphic robots





A new frontier: Humanoid Robots

- Putting together several types of robots we can create humanoid robots
-but this out of the scope of this course





Modelling: kinematics of manipulators

- **Kinematics** is about the motion of a robotic structure with respect to a generic frame
 - For a manipulator, given the joint position, find position and orientation of the end effector
- **Differential kinematics:**
 - For a manipulator: given the joint positions and motion (and their velocity) find the velocity of the end effector
- **Forward Kinematics:**
 - Find the end effector motion as a function of the joint motion
- **Inverse Kinematics**
 - Find the joint motion as a function of the “desired” end-effector motion and configuration



Modelling: kinematics of mobile robots

- Mobile robots:
 - The robot's kinematics requires a correct understanding of the motion constraints
 - The kinematic model translates the instantaneous motion of the motors into an instantaneous motion of the robot as a whole, which accounts for the robot's current configuration and for its motion constraints.